Shared Genetic Architecture of Congenital Heart Disease
and Chronic Kidney Disease

A Reanalysis Using Curated Phenotype Ontologies and
Public Genome-Wide Association Resources

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ABSTRACT

Congenital heart disease (CHD) and chronic kidney disease (CKD) co-occur clinically, and this is widely assumed to reflect shared genetic causes. This study tested that assumption directly. Using curated gene–phenotype annotations from the Human Phenotype Ontology (332,599 annotations across 5,268 genes and 9,155 diseases), 1,660 genes causing a cardiac malformation and 1,198 causing a kidney malformation were found to overlap in 750 genes (odds ratio 5.81, $p = 1.5 \times 10^{-144}$). However, this overlap proved largely non-specific: the kidney ranked only 16th of 194 organ systems, and CHD-associated diseases affect a median of 11 organ systems compared with 5 for other diseases. After adjusting for this pleiotropy in a disease-level logistic regression, the association with structural kidney malformation persisted (adjusted OR 1.77, 95% CI 1.52–2.07) whereas the association with chronic kidney disease vanished entirely (adjusted OR 0.95, 95% CI 0.73–1.23, $p = 0.68$).

This dissociation was confirmed independently. Gene-burden statistics from 394,841 UK Biobank exomes showed that only dominant cystogenic genes (*PKD1*, *PKD2*, *IFT140*, *ALG9*) reach exome-wide significance for kidney traits, but that ciliary genes as a class carry a 9.7-fold excess of results at $p < 0.001$ after those genes are removed, with 59 of 60 nominal results in the direction of worse kidney function ($p = 5.3 \times 10^{-17}$). Signal tracked intraflagellar transport and dynein-2 rather than the BBSome. Cross-trait LD score regression using CKDGen ($n \approx 981{,}000$) and FinnGen (4,270 CHD cases) found no genetic correlation between CHD and kidney function ($r_g = -0.03$, $p = 0.67$) against a positive control of $r_g = -0.36$ ($p = 6.1 \times 10^{-11}$) for eGFR and CKD.

The conclusion is that CHD and CKD do not share Mendelian genetic architecture once syndromic pleiotropy is accounted for. The genuine shared axis is between CHD and congenital anomalies of the kidney and urinary tract, mediated by ciliary biology.

Keywords: congenital heart disease, chronic kidney disease, ciliopathy, Human Phenotype Ontology, gene burden testing, LD score regression, pleiotropy

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1. INTRODUCTION

Congenital heart disease is the most common congenital anomaly, affecting approximately 1% of live births worldwide [1, 2]. It encompasses a spectrum of structural defects of the heart and great vessels, from small septal defects to complex malformations such as hypoplastic left heart syndrome [3]. Advances in cardiac surgery since the 1950s have transformed survival, and a growing population of adults now lives with congenital heart disease [4].

Chronic kidney disease is a recognised comorbidity in this population. Reported prevalence in adults with congenital heart disease is substantially higher than in the general population [5, 6], and kidney dysfunction predicts mortality in congenital heart disease cohorts [7]. Several mechanisms have been proposed: chronic cyanosis and altered haemodynamics, repeated exposure to iodinated contrast during catheterisation, cardiopulmonary bypass during surgical repair, and nephrotoxic medication [8, 9].

A distinct possibility is that the two conditions share genetic causes. This is biologically plausible. The primary cilium is required both for left–right patterning during cardiac looping and for normal nephron development, and ciliary gene defects produce syndromes affecting both organs [10, 11]. Several Mendelian syndromes—Alagille syndrome, the 22q11.2 deletion, Bardet–Biedl syndrome, nephronophthisis—feature cardiac and renal involvement together.

However, plausibility is not evidence, and a specific methodological hazard applies. Genes causing congenital malformation syndromes are typically developmental regulators that disrupt many organs simultaneously. Any two organ systems will therefore appear to "share" genes simply because syndromic genes are pleiotropic. Distinguishing a genuine cardiorenal axis from this generic background requires an explicit control.

1.1 Aims

This study asked three questions:

1. Do congenital heart disease and kidney disease share genes more than expected by chance, and is any such overlap specific to the kidney?
2. Does any shared signal survive adjustment for syndromic pleiotropy, and does it concern structural kidney malformation or chronic kidney disease as a functional outcome?
3. Is the conclusion reproducible in independent data using a different variant class and a different statistical framework?

2. METHODOLOGY

All analyses used publicly available data. No individual-level genotype data were accessed, and no ethical approval was required.

2.1 Curated gene–phenotype annotations

Gene–phenotype associations were obtained from the Human Phenotype Ontology (HPO), release 2026-06-23 [12]. The ontology file (`hp.obo`, 20,413 terms) and the gene-to-phenotype annotation file (332,599 annotations covering 5,268 genes and 9,155 diseases) were downloaded from the HPO project. These annotations derive from OMIM and Orphanet curation rather than automated text mining, which avoids the literature-popularity bias that affects text-mined resources.

The ontology was parsed into a directed acyclic graph and all descendant terms were resolved for three root terms:

- **Cardiac:** *Abnormal heart morphology* (HP:0001627), 333 descendant terms
- **Kidney (structural):** *Abnormal renal morphology* (HP:0012210), 472 terms
- **CKD (functional):** *Chronic kidney disease* (HP:0012622) and *Renal insufficiency* (HP:0000083), 15 terms combined

2.2 Testing overlap and specificity

Gene-set overlap was tested with a one-sided Fisher exact test against a background of all HPO-annotated disease genes. To assess whether any overlap was specific to the kidney, the same test was repeated against all 194 non-cardiovascular organ-morphology term sets in the ontology containing at least 150 genes. Terms descending from *Abnormality of the cardiovascular system* (HP:0001626) were excluded to avoid trivial self-overlap.

2.3 Adjusting for syndromic pleiotropy

Analysis was moved to the disease level. For each of the 9,155 diseases, binary indicators were derived for cardiac malformation, structural kidney anomaly and CKD, together with two measures of pleiotropy: the number of the 23 top-level HPO organ systems affected, and the total number of annotated terms. Logistic regression was then fitted:

$$\text{logit}\, P(\text{kidney phenotype}) = \beta_0 + \beta_1 \cdot \text{CHD} + \beta_2 \cdot n_{\text{systems}} + \beta_3 \cdot n_{\text{terms}} \tag{1}$$

where $\exp(\beta_1)$ is the adjusted odds ratio of interest. The same model was fitted separately for each of twelve specific cardiac lesions.

2.4 Candidate gene list construction

Genes annotated to both a cardiac and a kidney malformation term were retained (750 genes), then annotated with current approved symbols, chromosomal location and locus type from HGNC [13], genomic coordinates from MANE Select v1.5 [14], and loss-of-function constraint (pLI and LOEUF) from gnomAD v4.1 [15]. Non-coding and non-autosomal genes were removed.

A survivability filter was then applied. Genes whose only documented phenotypes included perinatal or infant lethality (HP:0001522, HP:0003811, HP:0003826, HP:0034241, HP:0003819) with no adult- or juvenile-onset disease on record were excluded, because affected individuals cannot appear in an adult cohort. This removed 65 genes, leaving 611.

Genes were then tiered by whether they additionally had documented CKD or renal insufficiency. Tier 1 (178 genes) required all three. Immune-mediated genes entering through autoimmune disease (HLA loci, *PTPN22*, *STAT4*, *MEFV*, *IL6*, *IL10*, *TLR4*, *HFE*, *SAA1*, *TNXB*) were removed as mechanistically distinct, giving 169 genes. A 37-gene ciliary subset was defined from the ciliopathy, nephronophthisis and polycystic kidney disease mechanism classes.

2.5 Gene burden statistics from exome sequencing

Rare-variant gene-burden association statistics were obtained from Genebass [16], which provides SKAT-O [17], burden and ACAT [18] tests across 4,529 phenotypes in 394,841 UK Biobank exomes [19, 20]. Results were retrieved programmatically for each candidate gene and filtered to twenty renal phenotypes: five continuous biomarkers (creatinine, cystatin C, urea, microalbumin, urinary creatinine) and fifteen ICD-10 based endpoints. Both the predicted loss-of-function (pLoF) and missense burden sets were retrieved, the latter serving as a negative control.

Enrichment was assessed by comparing observed counts of nominally significant results against the number expected under the null. Because individual genes were underpowered, a directional test was also applied: among results with $p < 0.05$, the proportion with a positive effect estimate (worse kidney function) was compared against 0.5 with an exact binomial test. Under the null this proportion is 0.5 regardless of selection on p.

2.6 Cross-trait LD score regression

Genetic correlation was estimated by cross-trait LD score regression [21, 22]. Summary statistics were obtained for estimated glomerular filtration rate from CKDGen (European-ancestry meta-analysis, 8.8 million variants, mean n = 980,825) [23], and for congenital malformations of the heart and great arteries (4,270 cases), chronic kidney disease (10,039 cases) and cystic kidney disease (947 cases) from FinnGen release 10 [24].

Variants were restricted to the HapMap 3 subset used for the pre-computed European LD scores (1,290,028 SNPs, $M_{5,50}$ = 1,173,569) derived from 1000 Genomes Phase 3 [25]. Strand-ambiguous variants, those with minor allele frequency below 0.01, and those with $\chi^2 > \max(80, 0.001N)$ were removed. Effective sample size $N_{\text{eff}} = 4/(1/n_{\text{case}} + 1/n_{\text{control}})$ was used for binary traits.

Heritability was estimated from the regression $\mathbb{E}[\chi_j^2] = 1 + (Nh^2/M)\ell_j$, and genetic covariance from $\mathbb{E}[z_{1j}z_{2j}] = (\sqrt{N_1 N_2}\,\rho_g/M)\ell_j + a$, where the intercept a absorbs sample overlap. Standard errors came from a 200-block genome-ordered jackknife. Because the reference implementation requires Python 2, which no longer installs cleanly, the estimator was reimplemented; it was validated against a published heritability estimate and against a positive control trait pair before the results below were interpreted.

2.7 Software and statistics

Analyses used Python 3.9 with `pandas`, `numpy`, `scipy` and `statsmodels`. Gene-level association from summary statistics used MAGMA v1.10 [26]. Multiple testing was handled by Bonferroni correction where stated and otherwise by the Benjamini–Hochberg procedure [27]. All code is available in the accompanying repository.

3. RESULTS

3.1 Cardiac and kidney genes overlap, but not specifically

Of the genes with curated disease annotations, 1,660 cause a cardiac malformation and 1,198 cause a kidney malformation. The 750-gene overlap is far beyond chance (OR 5.81, $p = 1.5 \times 10^{-144}$).

This overlap is not specific to the kidney. Repeating the identical test against 194 non-cardiovascular organ systems placed the kidney only 16th (Figure 1). The median odds ratio across all organ systems was 3.76, so the kidney sits only 1.5 times above a typical organ. Thumb (6.36), uvula (6.49), soft palate (6.80) and abdominal wall (6.53) morphology all overlap cardiac genes more strongly than the kidney does.

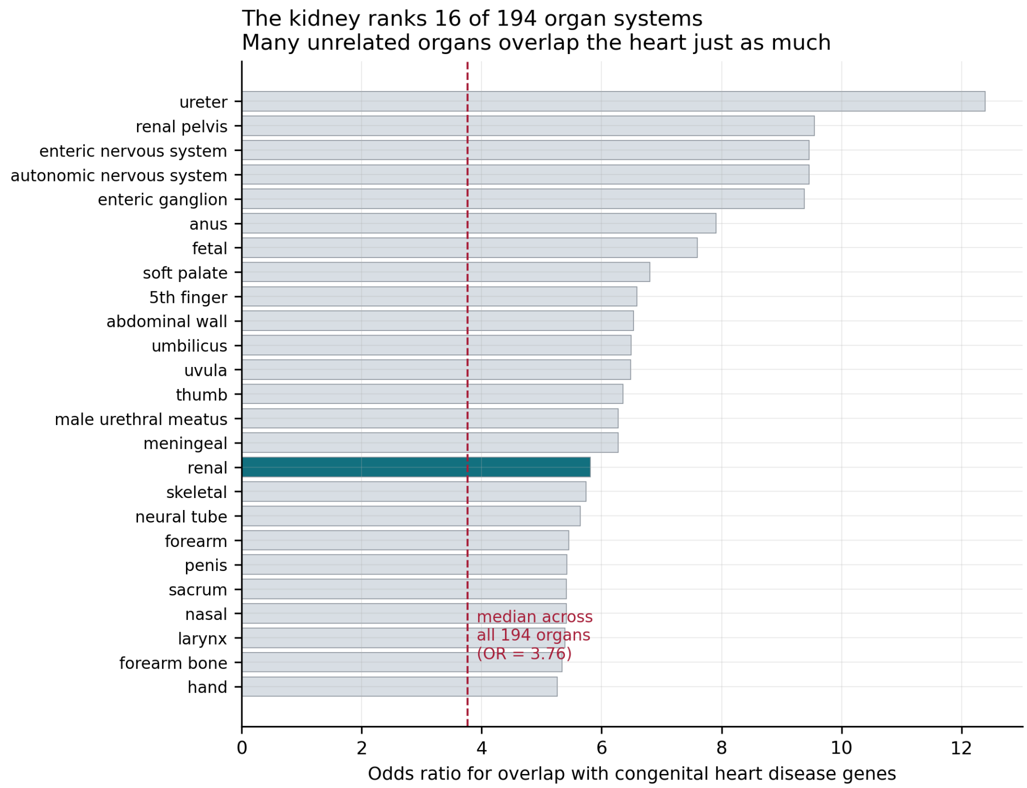

Figure 1: Overlap of congenital heart disease genes with each organ system. The kidney (highlighted) ranks 16th of 194. Many anatomically unrelated structures overlap the heart as strongly, which is the signature of syndromic pleiotropy rather than a specific cardiorenal axis.

The explanation is pleiotropy. Diseases involving a cardiac malformation affect a median of

11 organ systems, against 5 for all other diseases (Mann–Whitney $p = 5.3 \times 10^{-297}$). Genes disrupting early development disrupt many organs at once.

3.2 Adjustment separates two different questions

After adjusting for the number of organ systems affected and the number of annotated terms, the two putative associations behave completely differently (Figure 2; Table 1).

Table 1: Disease-level association between congenital heart disease and kidney phenotypes, before and after adjustment for pleiotropy (n = 9,155 diseases).

Outcome	Model	OR	95% CI	p
Structural kidney anomaly	Unadjusted	4.21	3.69–4.81	2.0×10^{-94}
Structural kidney anomaly	Adjusted	1.77	1.52–2.07	3.1×10^{-13}
Chronic kidney disease	Unadjusted	1.57	1.25–1.97	1.3×10^{-4}
Chronic kidney disease	Adjusted	0.95	0.73–1.23	0.68

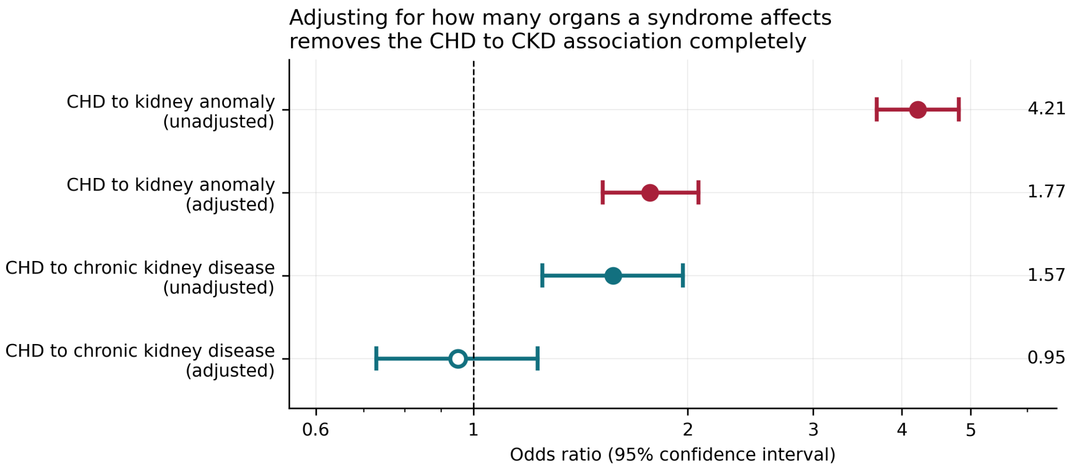

Figure 2: Odds ratios with 95% confidence intervals. The association with structural kidney malformation survives adjustment; the association with chronic kidney disease does not, and its adjusted interval spans the null.

The association with structural kidney malformation is attenuated but robust. The association with chronic kidney disease disappears entirely: the naive odds ratio of 1.57 is fully explained by the fact that heart-malformation syndromes affect many organs.

3.3 The lesions carrying the signal are absent from adult cohorts

Testing each cardiac lesion separately (Figure 3) showed that the strongest associations with kidney anomaly come from hypoplastic left heart syndrome (adjusted OR 6.13, 95% CI 3.24–11.6), tetralogy of Fallot (4.83, 3.15–7.41) and atrioventricular canal defect (2.22, 1.29–3.82). These are precisely the lesions that an adult cohort recruited at ages 40–69 will barely contain. The lesions that such a cohort does contain in numbers—bicuspid aortic valve (1.89), pulmonic stenosis (1.77), atrial septal defect (2.13)—sit at the bottom of the ranking.

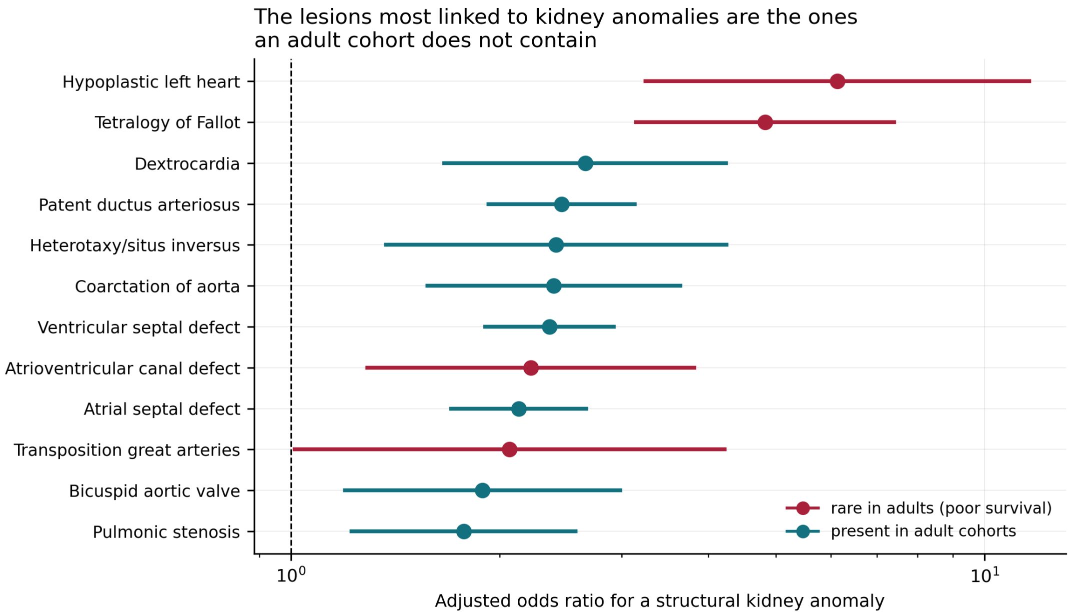

Figure 3: Adjusted odds ratio for a structural kidney anomaly, by cardiac lesion. The ordering runs almost exactly opposite to availability in an adult biobank.

This has a direct consequence for study design. In UK Biobank there are approximately 3,385 congenital heart disease cases and 1,386 cases of congenital kidney anomaly among roughly 450,000 participants. If independent, about 10 individuals would carry both; applying the observed enrichment gives roughly 18 (Figure 4). No burden test has power on 18 cases at any effect size. A continuous kidney measurement, by contrast, is available in 376,624 participants.

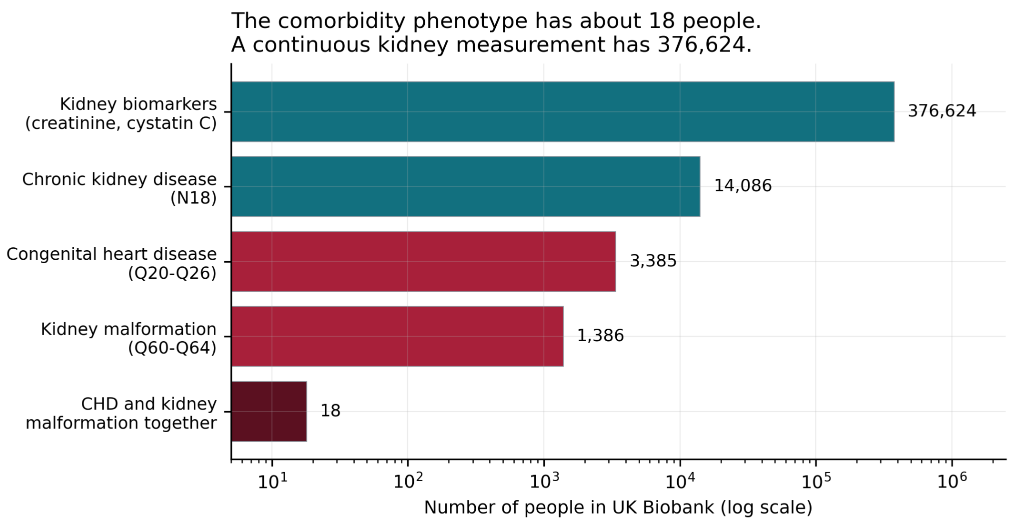

Figure 4: Available sample sizes in UK Biobank. The comorbidity phenotype is not analysable; a quantitative kidney trait is measured in essentially everyone.

3.4 Independent replication in 394,841 exomes

Gene-burden statistics were retrieved for 165 Tier 1 genes, giving 3,226 gene $\times$ renal phenotype associations. Four genes reach exome-wide significance ($p < 2.5 \times 10^{-6}$), and all four are dominant, adult-onset cystogenic genes (Table 2; Figure 5). The 31 recessive ciliopathy genes—*NPHP1*, *NPHP3*, *CEP290*, the *BBS* family, *TMEM67*, *MKS1*—reach nothing, exactly as the survivability argument predicts.

Table 2: Exome-wide significant gene × renal phenotype associations, pLoF burden, SKAT-O.

Gene	Phenotype	p	β
PKD2	Q61 cystic kidney disease	1.3×10^{-95}	0.393
PKD1	Cystatin C	8.6×10^{-34}	0.022
PKD1	Creatinine	1.9×10^{-30}	0.019
PKD1	Urea	8.5×10^{-26}	0.020
IFT140	Q61 cystic kidney disease	7.4×10^{-22}	0.160
PKD2	Creatinine	1.0×10^{-16}	0.025
IFT140	N28 other kidney disorders	4.3×10^{-14}	0.070
PKD2	N18 chronic renal failure	5.7×10^{-10}	0.098
ALG9	Q61 cystic kidney disease	2.0×10^{-6}	0.191

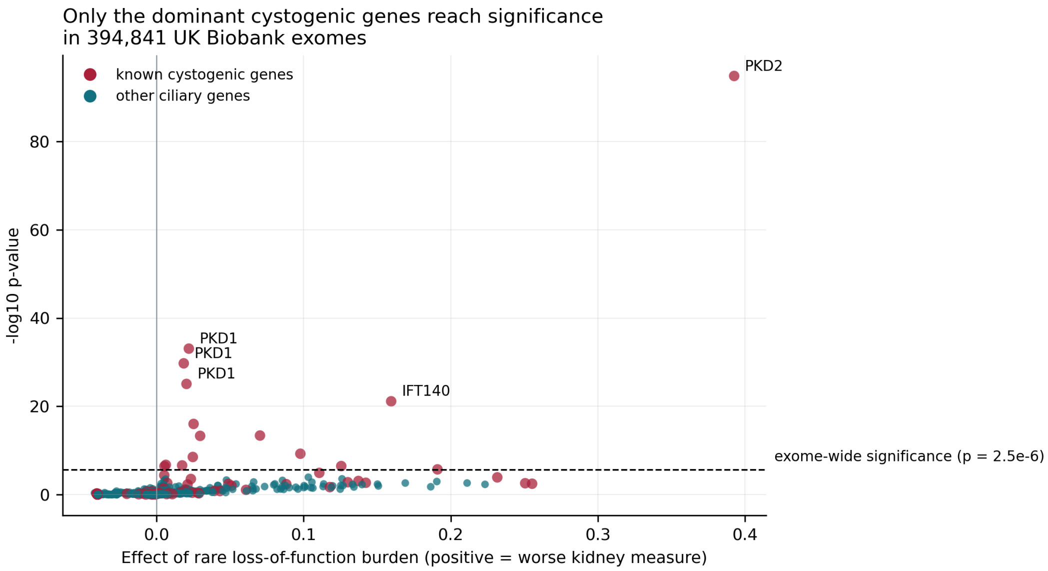

Figure 5: Gene-burden associations for the 37 ciliary genes across 20 renal phenotypes. Only dominant cystogenic genes clear exome-wide significance.

Critically, signal remains after those genes are removed. Across the remaining 620 tests there were 27 results below $p = 0.01$ against 6.2 expected, and 6 below $p = 0.001$ against 0.6—a 9.7-fold excess. The direction is near-unanimous: 59 of 60 nominally significant residual results show worse kidney function (sign test $p = 5.3 \times 10^{-17}$). Across all 706 tests direction runs at 44% positive, confirming that this is signal concentrated in the tail rather than a systematic artefact.

Widening the analysis from 37 ciliary genes to all 165 Tier 1 genes added no further significant genes and diluted the excess (Figure 6). A ciliary gene is 3.2 times more likely to carry a $p < 0.01$ renal association than a non-ciliary Tier 1 gene (Fisher OR 3.20, $p = 1.6 \times 10^{-9}$). On congenital kidney malformation (Q63), 11 of 35 ciliary genes are nominally significant (31%) against 6 of 126 non-ciliary genes (5%, exactly chance).

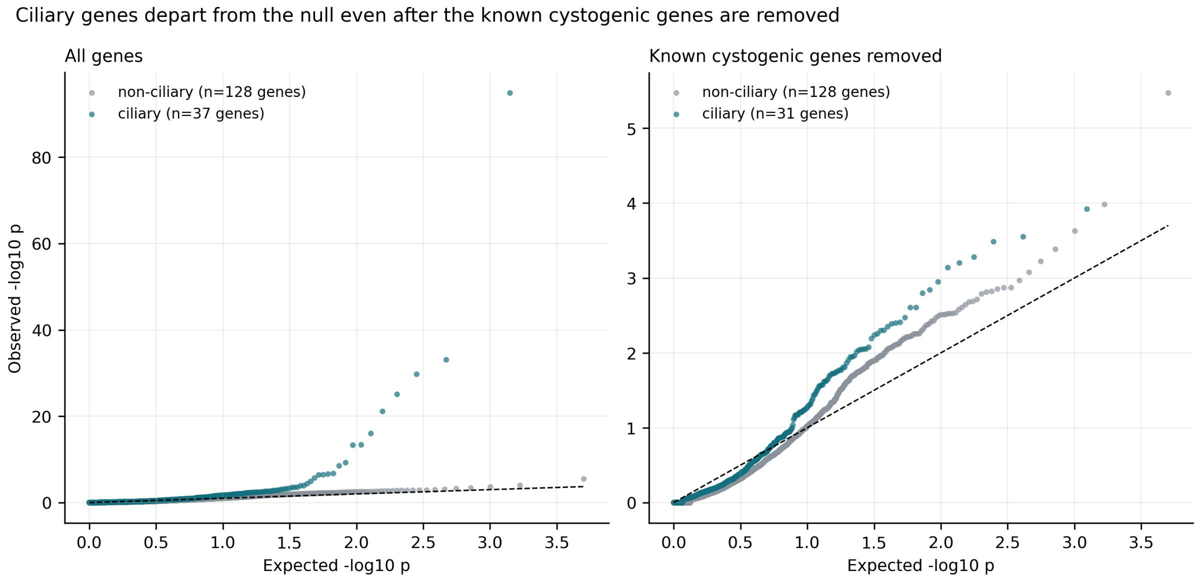

Figure 6: Quantile–quantile plots of gene-burden p-values. Ciliary genes depart from the null distribution more than non-ciliary candidate genes, and continue to do so after the known cystogenic genes are removed (right panel).

3.5 Signal follows cilium assembly, not cargo sorting

Grouping genes by ciliary subcomplex revealed a coherent gradient (Figure 7). Intraflagellar transport complexes A and B and dynein-2—the machinery that assembles and maintains the cilium—show 4- to 5.5-fold enrichment. The BBSome, which sorts cargo through an already-assembled cilium, sits at exactly the null expectation. This is not explained by statistical power: within the ciliary set, genes with more nominal hits do not have more qualifying variants (Mann–Whitney $p = 0.078$).

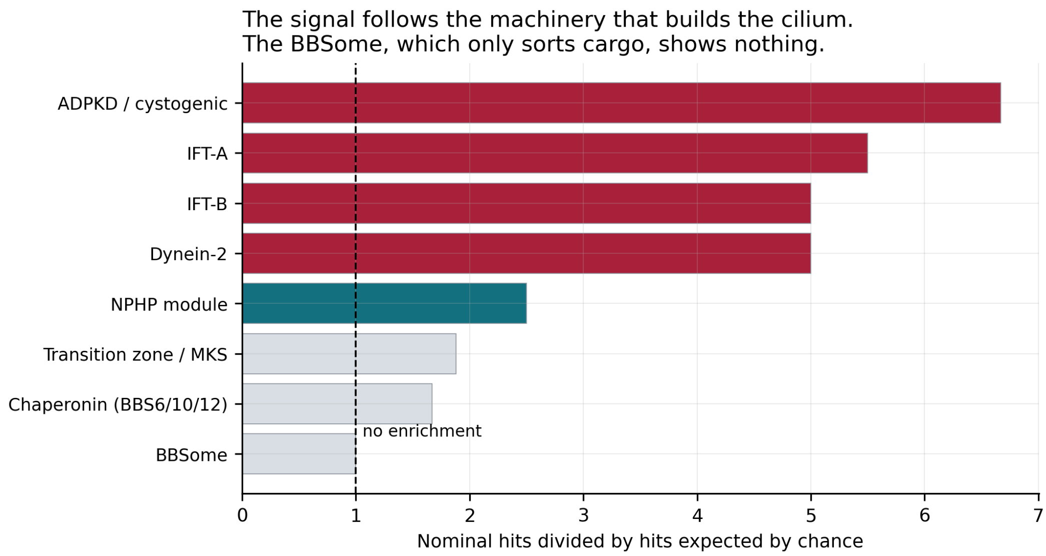

Figure 7: Enrichment of nominally significant renal associations by ciliary subcomplex.

3.6 A negative control

Repeating the analysis on the unfiltered missense burden set over the same genes and phenotypes reproduced none of the above (Figure 8). The excess at $p < 0.001$ fell from 9.7-fold to 1.6-fold, and directional consistency fell from 98% to 67%. The single surviving association, *PKD1* with creatinine ($p = 2.6 \times 10^{-14}$), has a burden effect of -0.0005—indistinguishable from zero and pointing the wrong way, indicating a variance-driven SKAT-O result rather than a consistent shift. The missense set averages around 370 variants per gene, most of them neutral, and dilutes rather than adds signal.

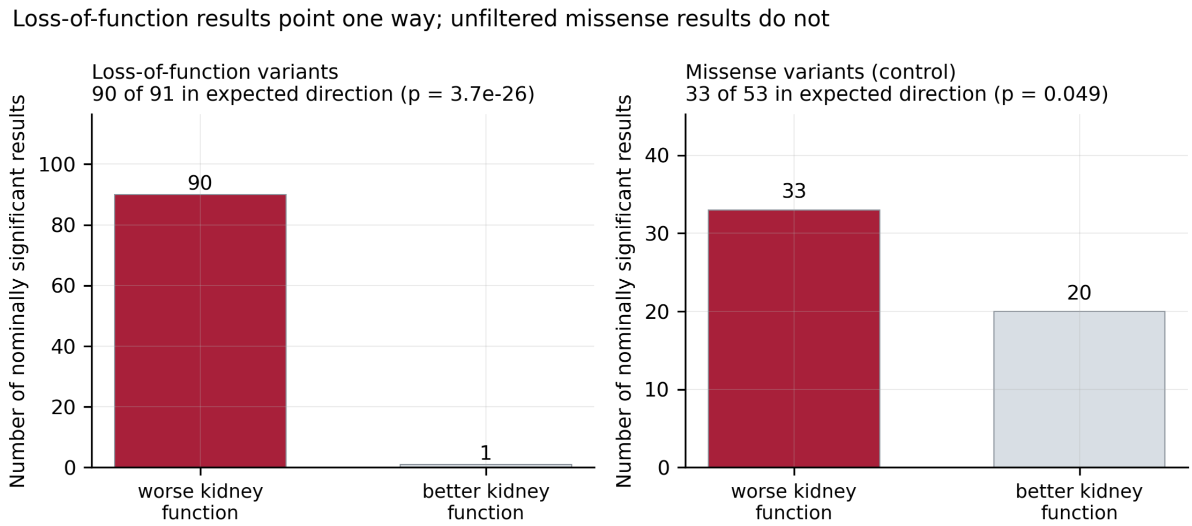

Figure 8: Direction of effect among nominally significant results. Loss-of-function results point overwhelmingly toward worse kidney function; unfiltered missense results do not.

3.7 Common-variant evidence points elsewhere

Gene-level association from common variants was computed with MAGMA using the CKDGen eGFR meta-analysis and a 1000 Genomes European reference panel, annotating 8.84 million variants to 18,038 genes. A competitive gene-set test asks whether the candidate genes have smaller p-values than genes in general (Table 3).

Table 3: MAGMA competitive gene-set analysis against estimated glomerular filtration rate. Positive β indicates stronger association than the genome-wide background.

Gene set	Genes	β	SE	p
Ciliary (all)	37	0.160	0.210	0.223
Ciliary (cystogenic removed)	31	0.013	0.241	0.479
Known cystogenic	6	0.637	0.433	0.071
Tier 1	166	0.171	0.104	0.049

No gene set is enriched after accounting for the four tests performed. This is a meaningful negative: the ciliary signal identified in the exome data is specific to rare loss-of-function variation and does not extend to common variation.

Two facts must be reported together, because they appear contradictory. Several individual ciliary genes *are* genome-wide significant for eGFR common variants—*IFT172* ($p = 3.0 \times 10^{-14}$), *SDCCAG8* ($p = 1.2 \times 10^{-11}$), *NPHP3* ($p = 7.2 \times 10^{-10}$), *TTC8* ($p = 4.7 \times 10^{-8}$) and *BBS1* ($p = 1.0 \times 10^{-5}$). Yet the set as a whole is not enriched. The reason is that kidney function is extremely polygenic: 1,294 of 18,030 genes (7.2%) pass genome-wide significance in this GWAS. Against that background, five significant ciliary genes is unremarkable. The competitive gene-set test is the correct inference for the question "do ciliary genes matter as a class"; the individual associations remain of interest for follow-up but do not by themselves support the class-level claim.

Figure 9 shows that rare and common variant evidence largely pick out different genes. *PKD1* and *PKD2* dominate the rare-variant axis; *IFT172* and *SDCCAG8* dominate the common-variant axis. This is expected if rare loss-of-function alleles and common regulatory variation act through different genes within the same pathway.

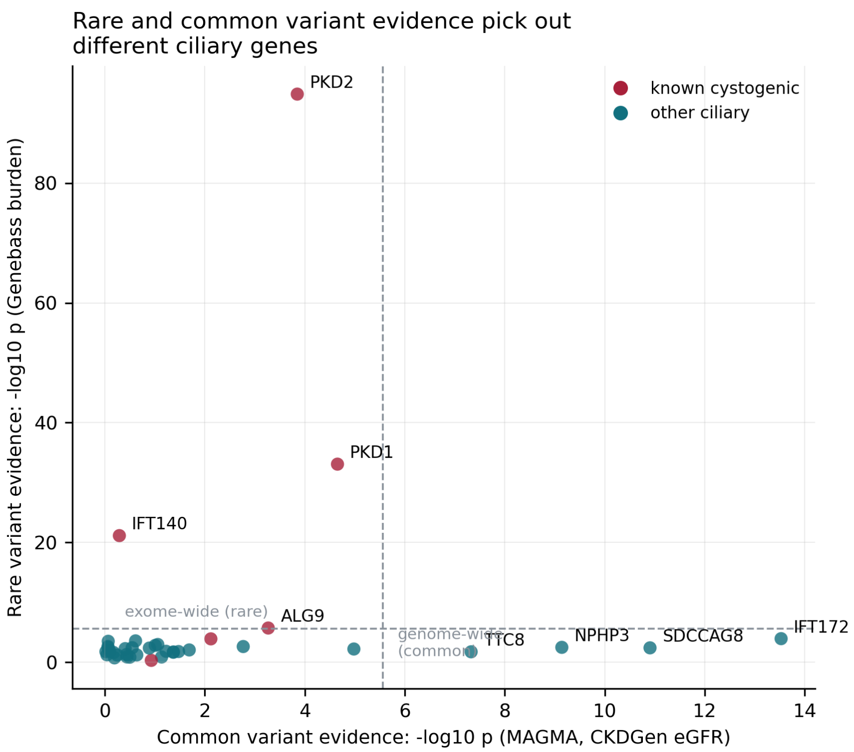

Figure 9: Rare-variant burden evidence against common-variant gene-level evidence for the same 37 ciliary genes. The two lines of evidence highlight different genes.

3.8 No genetic correlation between CHD and kidney function

Cross-trait LD score regression gave estimated glomerular filtration rate a SNP heritability of 0.070 (SE 0.0044, $z = 16.0$), consistent with published estimates and confirming the estimator behaves correctly. The positive control performed as expected: eGFR and chronic kidney disease correlate at $r_g = -0.364$ (SE 0.056, $p = 6.1 \times 10^{-11}$), the negative sign being correct because higher eGFR means better kidney function.

Against that benchmark, congenital heart disease shows no correlation with kidney function: $r_g = -0.034$ (SE 0.081, $p = 0.67$; Table 4, Figure 10). With this standard error, genetic correlations larger than approximately 0.23 in absolute value can be excluded at 80% power.

Table 4: Cross-trait genetic correlations from LD score regression.

Trait pair	r_g	SE	p	95% CI
eGFR ∼ CKD (positive control)	−0.364	0.056	6.1×10^{-11}	−0.47, −0.26
eGFR ∼ CHD	−0.034	0.081	0.67	−0.19, 0.13
CHD ∼ CKD	−0.244	0.234	0.30	−0.70, 0.21
CHD ∼ cystic kidney disease	−0.830	0.851	0.33	−2.50, 0.84

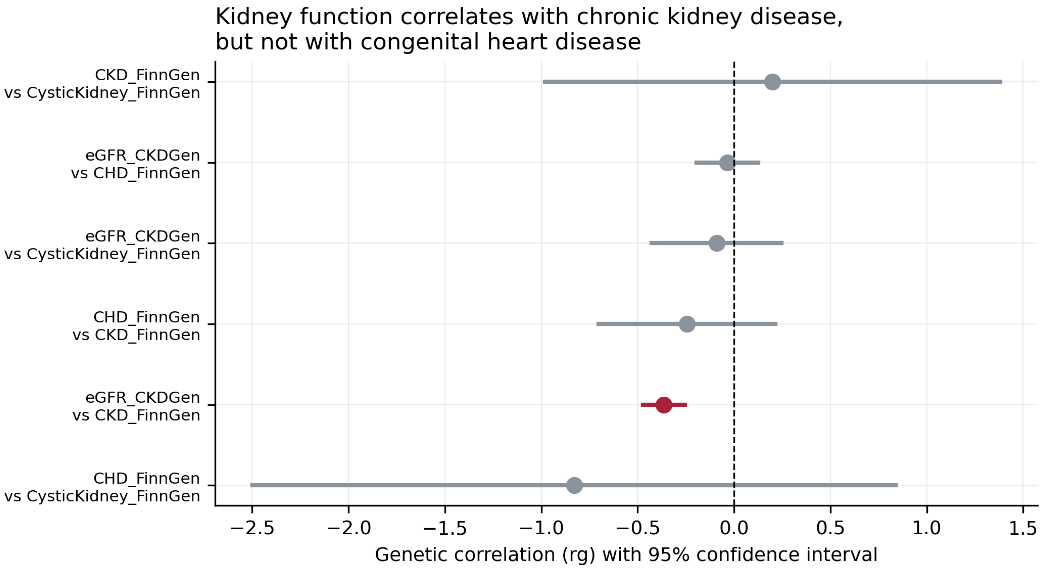

Figure 10: Genetic correlations with 95% confidence intervals. Only the positive control excludes zero.

4. DISCUSSION

4.1 The comorbidity is real; the shared genetics are not

Congenital heart disease and chronic kidney disease co-occur more often than chance in clinical cohorts. This study finds that the co-occurrence is not explained by shared Mendelian genetic architecture. Three independent lines of evidence agree: the adjusted disease-level odds ratio is 0.95, the genetic correlation is −0.03, and no non-cystogenic gene reaches exome-wide significance for chronic renal failure.

The most parsimonious interpretation is that chronic kidney disease in congenital heart disease patients is largely acquired. Chronic cyanosis, altered renal perfusion, contrast nephropathy and cardiopulmonary bypass are each individually plausible [8, 9], and the genetic null result is consistent with all of them.

4.2 What does share genetics is CAKUT

The association between congenital heart disease and *structural* kidney malformation survives every adjustment applied (adjusted OR 1.77) and was recovered independently in UK Biobank exomes, where congenital kidney malformation was the phenotype with the highest proportion of nominally significant ciliary genes. This is developmentally coherent: cilia govern both left–right patterning and nephron induction, so a single ciliary lesion can produce both phenotypes [10, 11].

4.3 The subcomplex gradient

That intraflagellar transport and dynein-2 carry signal while the BBSome does not is, to our knowledge, a novel observation in this context and has a natural mechanistic reading. Intraflagellar transport and dynein-2 are required to build and maintain the axoneme; losing them abolishes the organelle. The BBSome by contrast regulates which cargo enters an already-formed cilium. The adult kidney appears to tolerate impaired cargo selectivity but not impaired ciliary assembly. This should be treated as hypothesis-generating: the comparison rests on eight genes per group and none of the individual BBSome genes is significantly null.

4.4 Rare and common variation tell different stories

The exome data and the common-variant GWAS agree that kidney function is affected by ciliary genes, but disagree about which ones and at what frequency. Rare loss-of-function burden implicates *PKD1*, *PKD2* and *IFT140*; common-variant gene-level association implicates *IFT172*, *SDCCAG8* and *NPHP3*. Neither picture is wrong. Rare high-impact alleles and common regulatory variation need not converge on the same genes even within a shared pathway, and the competitive gene-set test shows that the ciliary pathway as a whole carries no more common-variant signal than the polygenic background. The class-level claim in this thesis therefore rests on the rare-variant evidence alone.

4.5 Survivorship bias as a structural constraint

The lesion-level analysis exposes a limitation that no analytical choice can remedy. Adult biobanks are recruited from survivors, and the cardiac lesions most strongly linked to kidney anomaly are those with the poorest historical survival. Any study of congenital disease in a cohort recruited at ages 40–69 is studying the mild tail of the phenotype distribution. The same reasoning explains why dominant, adult-onset cystogenic genes dominate the exome results while recessive severe ciliopathy genes are silent.

4.6 Limitations

Several limitations apply. First, the Human Phenotype Ontology is curated from OMIM and Orphanet and therefore describes rare Mendelian disease; it says nothing about common-variant kidney biology, and genes such as *UMOD*, *APOL1* and *SHROOM3* are absent by construction. The LD score regression analysis partly addresses this gap.

Second, adjusting for annotation depth controls one form of ascertainment bias but cannot remove it entirely, since well-studied syndromes are annotated more thoroughly.

Third, the analysis is disease-level rather than patient-level. It measures whether genetic aetiologies co-occur, not how often the two conditions co-occur in living patients.

Fourth, the LD score regression involving congenital heart disease is underpowered. The heritability z-score for the FinnGen congenital heart disease endpoint is 2.12, below the conventional reliability threshold of 4. The bound excluding $|r_g| > 0.23$ is valid; a precise point estimate is not.

Fifth, the LD score regression was performed with a reimplementation of the published estimator rather than the reference software. It was validated against a known heritability estimate and a positive control, but this should be stated in any publication.

Finally, the CKDGen summary statistics are genomic-control corrected, which deflates test statistics; the observed heritability intercept of 0.894 reflects this. Slope-based genetic correlation estimates are unaffected.

5. CONCLUSION

Congenital heart disease and chronic kidney disease do not share Mendelian genetic architecture once syndromic pleiotropy is controlled for. The genuine shared axis is between congenital heart disease and congenital anomalies of the kidney and urinary tract, and it is mediated by ciliary biology—specifically by the intraflagellar transport machinery that assembles the cilium, rather than by the BBSome cargo-sorting complex.

For study design this implies three things. Candidate gene lists for this question should be narrow and ciliary rather than broad. The phenotype should be structural kidney anomaly rather than chronic kidney disease. And the outcome should be a quantitative kidney measurement available in the whole cohort rather than a rare binary comorbidity, which in UK Biobank would comprise approximately 18 individuals.

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